Lctr 15: First/Second-Order Systems, Bode Plots, and Integrative Examples

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Focus: Sections 6.5–6.8 (Pages 456–490)

15.1 The Big Picture: Putting It All Together

Why This Matters

Lectures 12–14 built the Fourier transform machinery and used it to understand magnitude, phase, and ideal filters. This final lecture applies everything to **concrete systems**. Most systems encountered in practice—electrical circuits, mechanical structures, control loops—are described by first- or second-order differential equations (CT) or difference equations (DT). We will derive standard-form frequency responses for these systems, learn to sketch them quickly using **Bode plots**, and work through integrative examples that exercise the full analysis pipeline from Lecture 13.

Roadmap for this lecture:

1. First-order CT systems: standard form, time constant, Bode plot.
2. Bode plot conventions: asymptotic approximations, corner frequencies, dB/decade.
3. Second-order CT systems: natural frequency, damping ratio, resonance, step response.
4. First-order and second-order DT systems: pole radius and angle.
5. Integrative worked examples: full pipeline from equation to output.

15.2 First-Order CT Systems

15.2.1 Standard Form

In Lecture 11, we derived the frequency response of a specific RC circuit. Now we generalize. *Any* first-order CT LTI system governed by

$$\frac{dy}{dt} + \omega_c\, y(t) = \omega_c\, x(t)$$

has the frequency response (replacing $d/dt \to j\omega$, as in Lecture 13):

$$\boxed{H(j\omega) = \frac{\omega_c}{j\omega + \omega_c} = \frac{1}{1 + j\omega/\omega_c}}$$

This is the **standard first-order lowpass** response. The single parameter ω_c (in rad/s) determines everything.

15.2.2 Key Characteristics

Time constant: $\tau = 1/\omega_c$. The impulse response is $h(t) = \omega_c\, e^{-\omega_c t}\, u(t)$, which decays to $1/e$ of its initial value in time τ.

Bandwidth: The -3 dB frequency is ω_c (by definition of the standard form). At $\omega = \omega_c$: $|H| = 1/\sqrt{2} \approx 0.707$, meaning the power is halved.

DC gain: $H(0) = 1$ (unity gain at zero frequency).

Magnitude:

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega/\omega_c)^2}}, \qquad |H|_{\text{dB}} = -10\log_{10}\big(1 + (\omega/\omega_c)^2\big)$$

Phase:

$$\angle H(j\omega) = -\arctan(\omega/\omega_c)$$

First-Order System with Cutoff at 100 rad/s

Let $\omega_c = 100$ rad/s (equivalently, $f_c = 100/(2\pi) \approx 15.9$ Hz, $\tau = 10$ ms).

Magnitude at key frequencies:

- $\omega = 10$: $|H| = 1/\sqrt{1.01} = 0.995$ (-0.04 dB) — essentially passed.
- $\omega = 100 = \omega_c$: $|H| = 1/\sqrt{2} = 0.707$ (-3 dB) — cutoff.
- $\omega = 1000 = 10\omega_c$: $|H| = 1/\sqrt{101} = 0.0995$ (-20 dB) — strongly attenuated.
- $\omega = 10000 = 100\omega_c$: $|H| = 1/\sqrt{10001} = 0.01$ (-40 dB).

Phase at key frequencies:

- $\omega = 10$: $\angle H = -\arctan(0.1) = -5.7°$ — nearly zero.
- $\omega = 100$: $\angle H = -\arctan(1) = -45°$ — the halfway point.
- $\omega = 1000$: $\angle H = -\arctan(10) = -84.3°$ — approaching $-90°$.

15.3 Bode Plots

Why This Matters

A **Bode plot** consists of two graphs: magnitude (in dB) and phase (in degrees or radians), both plotted against **frequency on a logarithmic scale**. The logarithmic frequency axis is essential: it compresses wide frequency ranges and reveals the asymptotic behavior of the response as straight lines. Bode plots are the standard visualization tool for frequency responses in electrical engineering, control systems, and communications.

15.3.1 Bode Plot of the First-Order System

The magnitude in dB is:

$$|H|_{\text{dB}} = -10\log_{10}\big(1 + (\omega/\omega_c)^2\big)$$

Low-frequency asymptote ($\omega \ll \omega_c$): $|H|_{\text{dB}} \approx -10\log_{10}(1) = 0$ dB. (Flat.)

High-frequency asymptote ($\omega \gg \omega_c$): $|H|_{\text{dB}} \approx -10\log_{10}(\omega/\omega_c)^2 = -20\log_{10}(\omega/\omega_c)$.

This decreases by 20 dB for every factor-of-10 increase in frequency, called a **slope of** -20 **dB/decade** (or equivalently -6 dB/octave).

The two asymptotes meet at $\omega = \omega_c$ — the **corner frequency** (or **break frequency**).

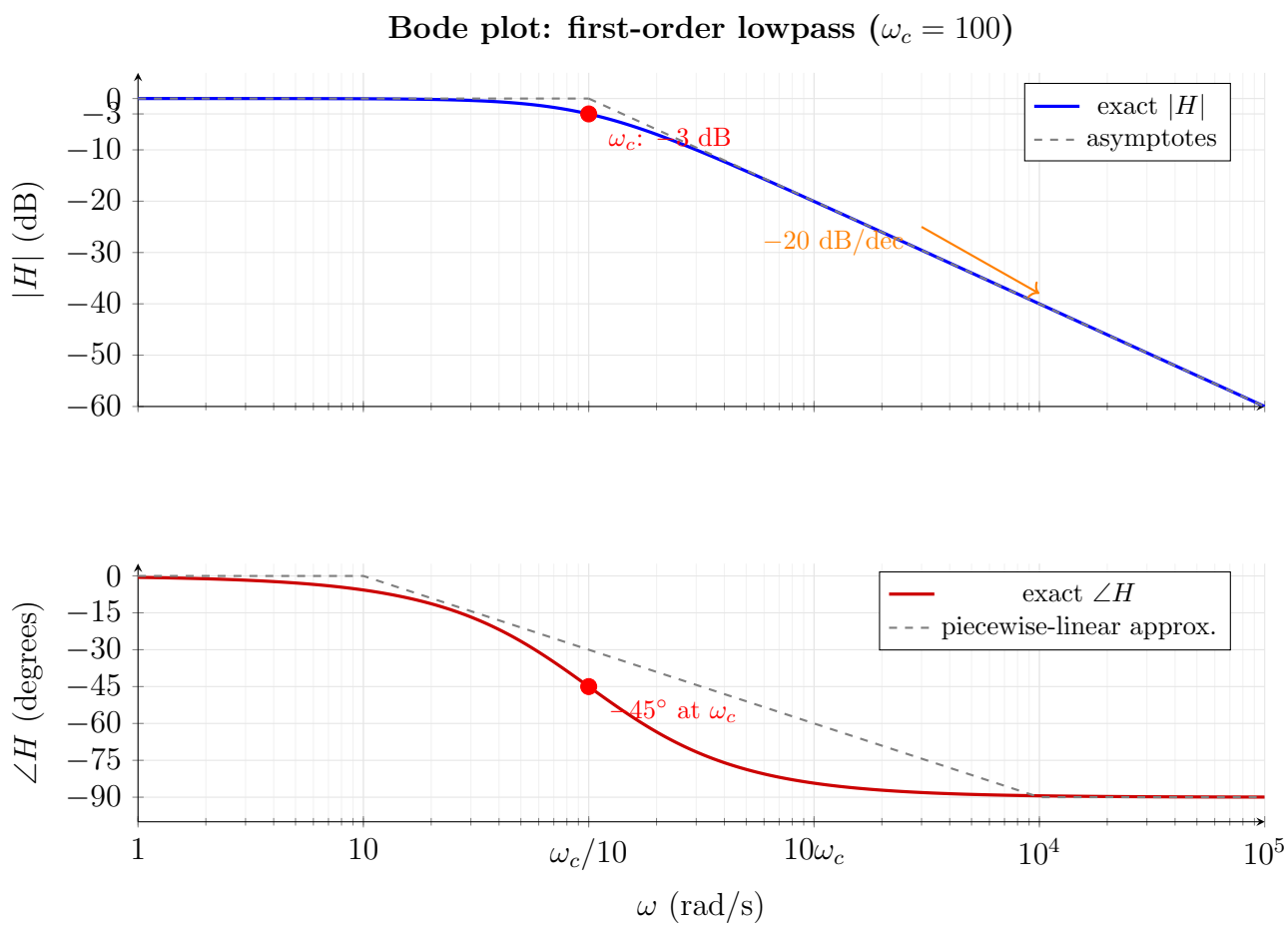

Figure 1: Bode plot of a first-order lowpass system with $\omega_c = 100$ rad/s. Top: Magnitude rolls off at -20 dB/decade above the corner frequency. Bottom: Phase transitions from 0° (low freq.) to $-90°$ (high freq.), passing through $-45°$ at ω_c. The dashed lines show the asymptotic/piecewise-linear approximations used for quick hand-sketching.

15.3.2 Bode Plot Rules for Hand Sketching

Key Insight

Quick-sketch rules for a first-order pole at ω_c:

1. **Magnitude:** Draw 0 dB for $\omega < \omega_c$, then a line with slope -20 dB/decade for $\omega > \omega_c$. The true curve is 3 dB below the asymptote at ω_c, and 1 dB below at $\omega_c/2$ and $2\omega_c$.

2. **Phase:** Draw $0°$ for $\omega < \omega_c/10$, then a line from $0°$ to $-90°$ spanning one decade on each side of ω_c (i.e., from $\omega_c/10$ to $10\omega_c$), then $-90°$ above $10\omega_c$. The true value is $-45°$ at ω_c.

For a first-order **zero** at ω_z: same shapes but with **positive** slopes (+20 dB/decade, phase goes from $0°$ to $+90°$).

15.3.3 First-Order Highpass

The complementary first-order highpass has:

$$H_{\text{HP}}(j\omega) = \frac{j\omega/\omega_c}{1 + j\omega/\omega_c} = 1 - H_{\text{LP}}(j\omega)$$

Its Bode plot is the mirror image: magnitude rises at +20 dB/decade below ω_c and flattens at 0 dB above ω_c. Phase goes from $+90°$ (low freq.) to $0°$ (high freq.).

15.4 Second-Order CT Systems

Why This Matters

Second-order systems are far richer than first-order ones. They exhibit **resonance**, **oscillatory step responses**, and a range of behaviors (underdamped, critically damped, overdamped) controlled by a single parameter: the **damping ratio** ζ. Almost every physical system with energy storage in two forms (e.g., an inductor and a capacitor, or a spring and a mass) is governed by a second-order equation.

15.4.1 Standard Form

A general second-order system is governed by:

$$\frac{d^2y}{dt^2} + 2\zeta\omega_n \frac{dy}{dt} + \omega_n^2\, y(t) = \omega_n^2\, x(t)$$

where ω_n is the **natural frequency** (rad/s) and ζ is the **damping ratio** (dimensionless). Replacing $d/dt \to j\omega$:

$$\boxed{H(j\omega) = \frac{\omega_n^2}{(j\omega)^2 + 2\zeta\omega_n(j\omega) + \omega_n^2} = \frac{1}{1 - (\omega/\omega_n)^2 + j\,2\zeta(\omega/\omega_n)}}$$

15.4.2 Magnitude Response

Let $u = \omega/\omega_n$ (normalized frequency). Then:

$$|H(ju\omega_n)| = \frac{1}{\sqrt{(1-u^2)^2 + (2\zeta u)^2}}$$

Key values:

- At $\omega = 0$ ($u = 0$): $|H| = 1$ (DC gain is unity).
- At $\omega = \omega_n$ ($u = 1$): $|H| = 1/(2\zeta)$. For $\zeta < 0.5$, this exceeds unity—there is amplification at ω_n.
- The **resonance peak** occurs at:

  $$\omega_r = \omega_n\sqrt{1 - 2\zeta^2} \qquad \text{(exists only when } \zeta < 1/\sqrt{2} \approx 0.707\text{)}$$

  with peak magnitude:

  $$|H|_{\max} = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

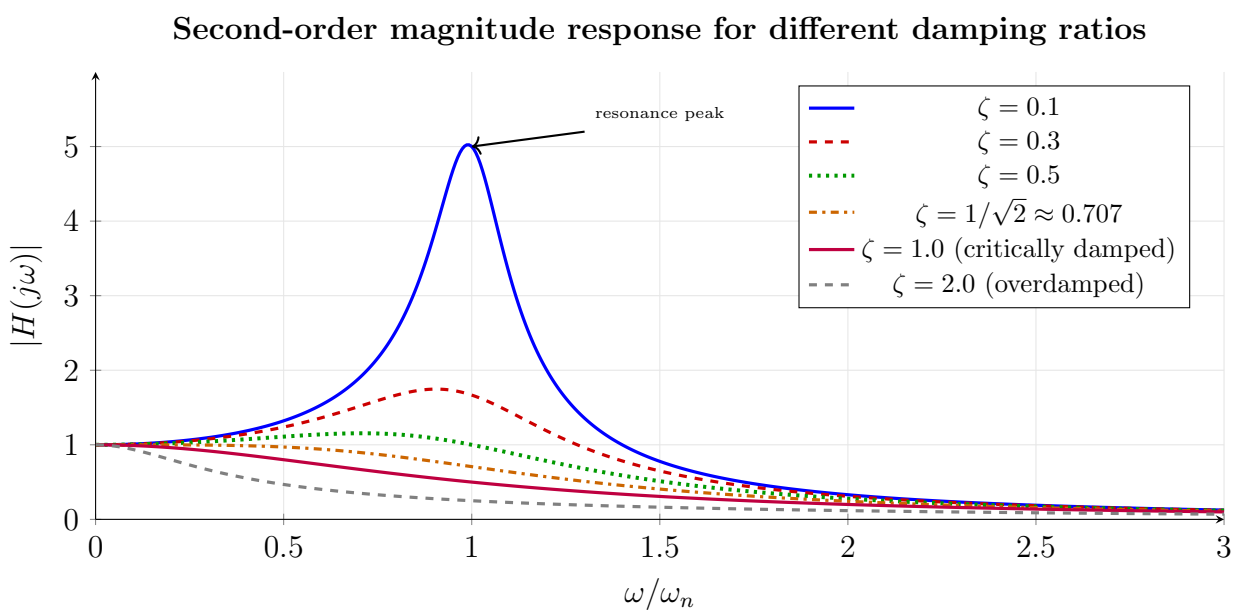

Figure 2: Second-order magnitude response. Smaller ζ produces a taller, sharper resonance peak near ω_n. At $\zeta = 1/\sqrt{2}$, the response is "maximally flat" (Butterworth)—no peaking. For $\zeta \geq 1$ (critically damped or overdamped), the response rolls off monotonically without any peak.

15.4.3 Phase Response

$$\angle H = -\arctan\left(\frac{2\zeta\,(\omega/\omega_n)}{1-(\omega/\omega_n)^2}\right)$$

Key values:

- At $\omega = 0$: phase $= 0°$.
- At $\omega = \omega_n$: phase $= -90°$ **for all values of** ζ. (The natural frequency is always the $-90°$ point.)
- At $\omega \to \infty$: phase $\to -180°$.

The phase transition is sharper for smaller ζ (a narrow resonance produces a rapid phase swing).

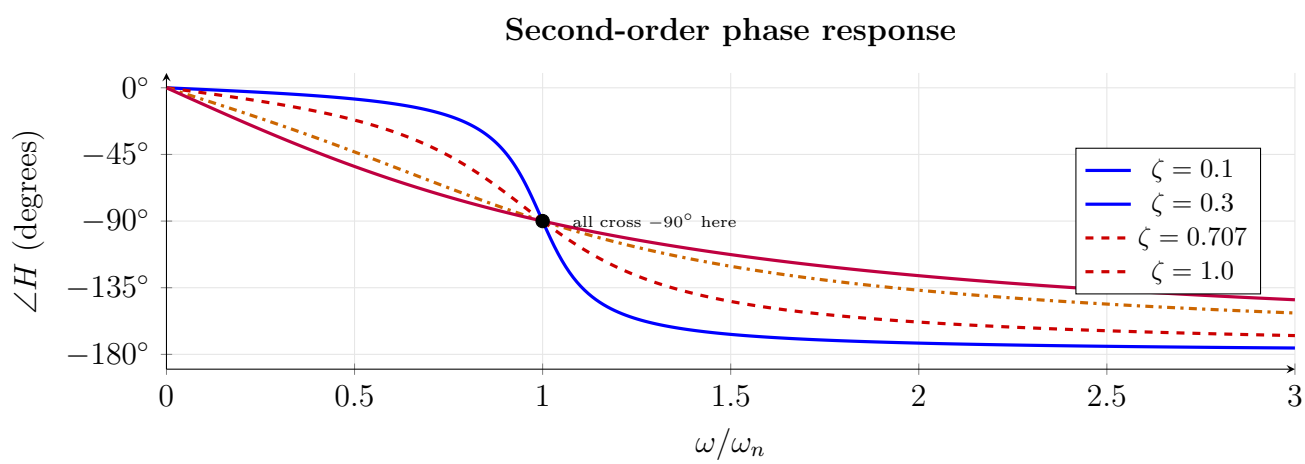

Figure 3: All second-order systems pass through $-90°$ at $\omega = \omega_n$, regardless of ζ. Lightly damped systems ($\zeta \ll 1$) have an abrupt phase transition; heavily damped systems have a gradual one.

15.4.4 Step Response and Transient Behavior

The step response of the standard second-order system reveals the time-domain consequences of the damping ratio. For the **underdamped** case ($\zeta < 1$):

$$s(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin\!\big(\omega_d\, t + \phi\big)\, u(t), \qquad \omega_d = \omega_n\sqrt{1-\zeta^2}, \quad \phi = \arccos(\zeta)$$

where ω_d is the **damped natural frequency**.

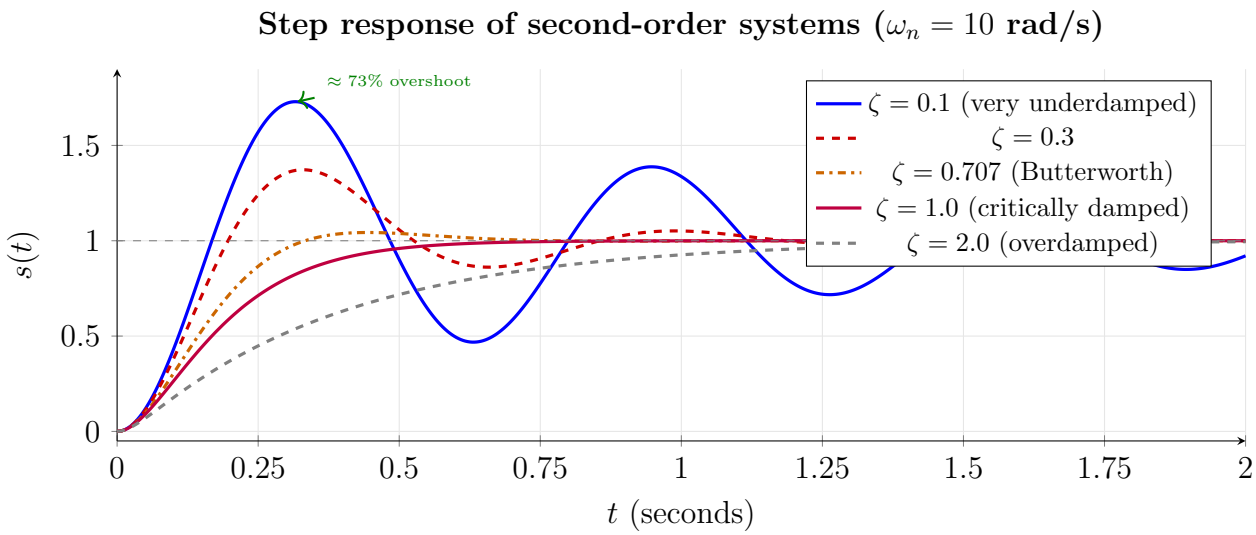

Figure 4: Step responses for different damping ratios. Smaller ζ produces more oscillation and overshoot but faster initial response. The critically damped case ($\zeta = 1$) is the fastest response *without* overshoot. The Butterworth case ($\zeta = 1/\sqrt{2}$) has slight overshoot ($\approx 4\%$) but offers the flattest magnitude response.

Key Insight

Step response specifications:

- **Percent overshoot:** $\%\text{OS} = 100 \exp\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right)$. For $\zeta = 0.1$: 73%. For $\zeta = 0.5$: 16%. For $\zeta = 0.707$: 4.3%.
- **Settling time** (to within 2% of final value): $t_s \approx 4/(\zeta\omega_n)$.
- **Peak time:** $t_p = \pi/\omega_d = \pi/(\omega_n\sqrt{1-\zeta^2})$.

These formulas are heavily used in control systems design.

15.4.5 Bode Plot of the Second-Order System

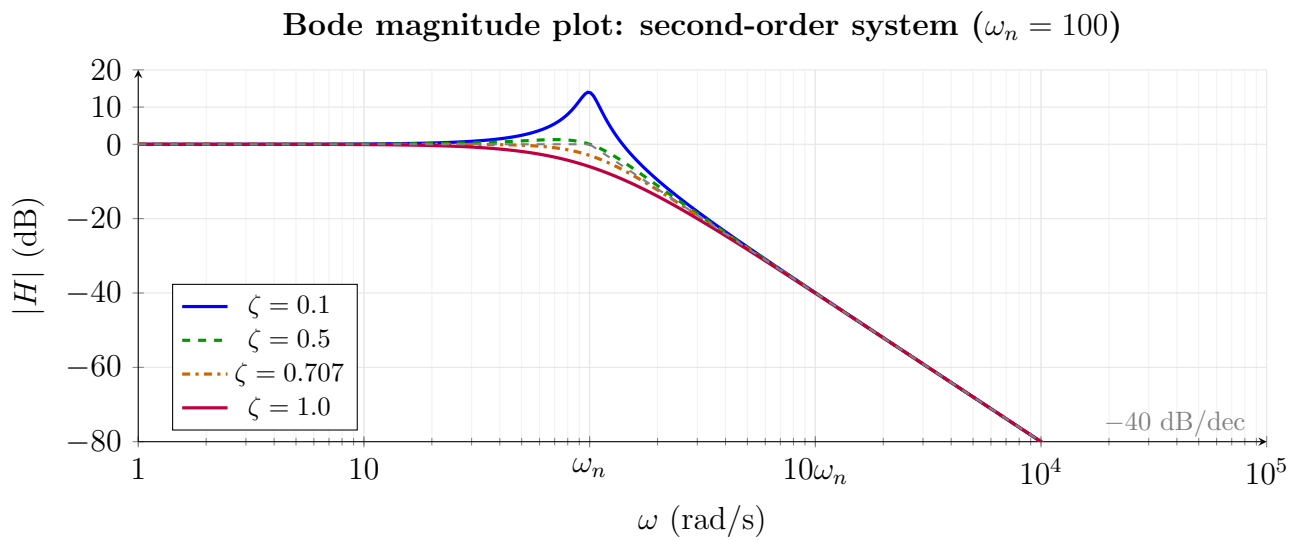

Figure 5: Second-order Bode magnitude plot. Below ω_n: flat at 0 dB. Above ω_n: rolls off at -40 dB/decade (twice the first-order rate). Near ω_n, the peak depends on ζ: smaller ζ gives a taller peak.

Key Insight

Bode plot rules for a second-order system:

- High-frequency asymptote: -40 dB/decade (two poles $\Rightarrow$ twice the slope).
- The actual curve deviates from the asymptote near ω_n by an amount that depends on ζ. For $\zeta = 0.707$, the deviation is $+3$ dB; for $\zeta = 0.1$, the peak is $+14$ dB above the asymptote.
- Phase goes from $0°$ to $-180°$, with $-90°$ always at ω_n.

15.5 First-Order and Second-Order DT Systems

Why This Matters

DT systems follow the same structural pattern as CT systems, but with poles inside the **unit circle** (instead of the left half-plane) and frequency response evaluated on the unit circle $z = e^{j\omega}$ (instead of the $j\omega$ axis). The parallels are exact; only the geometry changes.

15.5.1 First-Order DT System

The first-order DT lowpass system (from Lecture 11):

$$y[n] = ay[n-1] + (1-a)x[n], \quad 0 < a < 1$$

has frequency response:

$$H(e^{j\omega}) = \frac{1-a}{1-ae^{-j\omega}}$$

This has a single **pole at** $z = a$ (inside the unit circle for stability).

Magnitude:

$$|H(e^{j\omega})| = \frac{1-a}{\sqrt{1+a^2-2a\cos\omega}}$$

At $\omega = 0$: $|H| = (1-a)/(1-a) = 1$ (DC gain unity, by design). At $\omega = \pi$: $|H| = (1-a)/(1+a)$, which is small for a close to 1 (lowpass behavior).

Comparing CT and DT First-Order Systems

CT: pole at $s = -\omega_c$. $|H|$ drops to $1/\sqrt{2}$ at $\omega = \omega_c$.
DT: pole at $z = a$. The -3 dB frequency depends on a:

$$\omega_{-3\text{dB}} = \arccos\left(\frac{1+a^2-\left(\frac{1-a}{1/\sqrt{2}}\right)^2}{2a}\right) = \arccos\left(\frac{2a^2-a^2+2a-1}{2a}\right)$$

For $a = 0.9$: $\omega_{-3\text{dB}} \approx 0.325$ rad/sample $\approx \pi/10$. The closer a is to 1, the narrower the passband (stronger lowpass effect).

15.5.2 Second-Order DT System

A second-order DT system with complex conjugate poles at $z = re^{\pm j\theta}$ (where $r < 1$ for stability) has:

$$H(e^{j\omega}) = \frac{G}{(1-re^{j\theta}e^{-j\omega})(1-re^{-j\theta}e^{-j\omega})}$$

where $G = 1 + r^2 - 2r\cos\theta$ normalizes the DC gain to unity ($|H(e^{j0})| = 1$).

DT–CT parallel:

Parameter	**CT (poles at $s = -\sigma \pm j\omega_d$)**	**DT (poles at $z = re^{\pm j\theta}$)**
Stability	$\sigma > 0$ (left half-plane)	$r < 1$ (inside unit circle)
Damping	$\zeta = \sigma/\omega_n$	$r = e^{-\sigma T} \approx 1 - \sigma T$
Resonance freq.	$\omega_r \approx \omega_n$	$\omega_r \approx \theta$
More damping	Larger ζ	Smaller r
Oscillation freq.	$\omega_d = \omega_n\sqrt{1-\zeta^2}$	θ

The pole **radius** r controls damping (smaller r = more damped), and the pole **angle** θ controls the resonance/oscillation frequency.

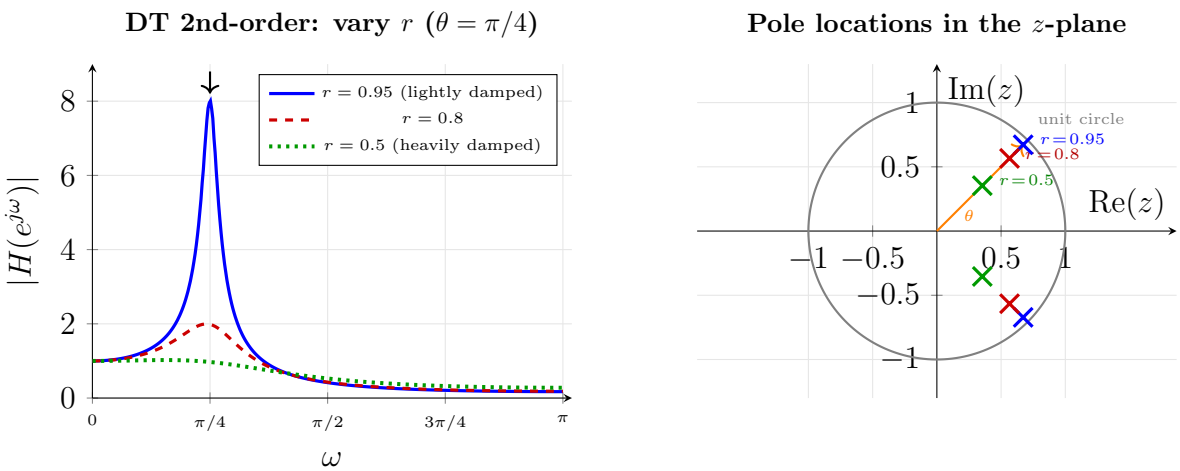

Figure 6: Left: DT second-order magnitude responses for poles at angle $\theta = \pi/4$ with different radii. Poles closer to the unit circle (larger r) produce sharper resonance peaks. Right: The corresponding pole locations in the z-plane. All poles are at angle $\theta = \pi/4$; only the radius differs.

15.6 Integrative Worked Examples

Why This Matters

The following examples exercise the complete analysis pipeline developed in Lectures 12–15: starting from a system description, finding the frequency response, characterizing the filter behavior, and computing the output for a given input.

15.6.1 CT Example: Automobile Suspension System

Automobile Suspension (CT Second-Order)

An automobile suspension can be modeled as a second-order system where the input $x(t)$ is the road surface profile and the output $y(t)$ is the vertical displacement of the car body. The governing equation is:

$$m\ddot{y} + b\dot{y} + ky = b\dot{x} + kx$$

where $m = 500$ kg, $b = 2000$ N·s/m, $k = 20000$ N/m.
Step 1: Find the frequency response. Replace $d/dt \to j\omega$:

$$H(j\omega) = \frac{b(j\omega) + k}{m(j\omega)^2 + b(j\omega) + k} = \frac{j2000\omega + 20000}{-500\omega^2 + j2000\omega + 20000}$$

Dividing numerator and denominator by $k = 20000$:

$$H(j\omega) = \frac{1 + j\omega/10}{1 - \omega^2/40 + j\omega/10}$$

Step 2: Identify parameters. The denominator has the standard form with:

$$\omega_n = \sqrt{k/m} = \sqrt{40} \approx 6.32 \text{ rad/s}, \qquad \zeta = \frac{b}{2\sqrt{mk}} = \frac{2000}{2\sqrt{10^7}} \approx 0.316$$

This is **underdamped** ($\zeta < 1$), so the car will oscillate after hitting a bump.
Step 3: Characterize the behavior.

- DC ($\omega = 0$): $H(0) = 1$ — the car follows constant road elevation changes perfectly.
- Near $\omega_n \approx 6.3$ rad/s: the system resonates. The magnitude peaks.
- High frequency ($\omega \gg \omega_n$): $|H| \to 0$ — rapid bumps are filtered out. The suspension acts as a lowpass filter.

Step 4: Compute the resonance peak. At $\omega = \omega_n$:

$$|H(j\omega_n)| = \frac{|1 + j\omega_n/10|}{|j\omega_n/10| \cdot 2\zeta} \cdot \frac{1}{1} \approx \frac{\sqrt{1 + (6.32/10)^2}}{2(0.316)} \approx \frac{1.183}{0.632} \approx 1.87$$

The car body oscillates at nearly **twice** the road disturbance amplitude at the resonance frequency—this is why poor suspension design leads to a bouncy ride.

15.6.2 DT Example: Moving-Average Filter Design

Designing a DT Lowpass Filter

Goal: Design a simple FIR lowpass filter to smooth noisy sensor data sampled at $f_s = 1000$ Hz. We want to pass frequencies below 50 Hz and attenuate above 100 Hz.
Step 1: Convert to normalized frequency. $\omega = 2\pi f/f_s$, so:

- Passband edge: $\omega_p = 2\pi(50)/1000 = 0.1\pi$ rad/sample.
- Stopband edge: $\omega_s = 2\pi(100)/1000 = 0.2\pi$ rad/sample.
- Transition bandwidth: $\Delta\omega = 0.1\pi$.

Step 2: Choose filter length. A rule of thumb for a rectangular-windowed FIR filter: $M \approx 4\pi/\Delta\omega = 4\pi/(0.1\pi) = 40$. So we use $M = 40$ (a 41-tap filter).
Step 3: Design coefficients. The ideal lowpass impulse response truncated to 41 taps:

$$h[n] = \frac{\sin\big(\omega_c(n-20)\big)}{\pi(n-20)}, \quad n = 0, 1, \ldots, 40$$

with $\omega_c = 0.15\pi$ (midpoint between ω_p and ω_s) and $h[20] = \omega_c/\pi = 0.15$.
Step 4: Verify. The frequency response is $H(e^{j\omega}) = \sum_{n=0}^{40} h[n]\, e^{-j\omega n}$. This is a linear-phase FIR filter (symmetric coefficients) with group delay = 20 samples = 20 ms.
Performance: Passband ripple ≈ 0.5 dB (due to Gibbs phenomenon from rectangular truncation). Stopband attenuation ≈ 21 dB. Better performance requires windowing (Hamming, Hanning) or more taps.

15.6.3 Full-Pipeline CT Example

From Equation to Output: Complete Pipeline

System: $\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 4y = 2x(t)$
Input: $x(t) = e^{-t}u(t)$, so $X(j\omega) = \frac{1}{1+j\omega}$.
Step 1: Frequency response.

$$H(j\omega) = \frac{2}{(j\omega)^2 + 5(j\omega) + 4} = \frac{2}{(j\omega+1)(j\omega+4)}$$

Step 2: Output spectrum.

$$Y(j\omega) = X \cdot H = \frac{1}{1+j\omega} \cdot \frac{2}{(j\omega+1)(j\omega+4)} = \frac{2}{(1+j\omega)^2(4+j\omega)}$$

Step 3: Partial fractions.

$$\frac{2}{(1+j\omega)^2(4+j\omega)} = \frac{A}{1+j\omega} + \frac{B}{(1+j\omega)^2} + \frac{C}{4+j\omega}$$

Multiplying through and solving: set $j\omega = -1$: $B(3) = 2 \Rightarrow B = 2/3$.
Set $j\omega = -4$: $C(9) = 2 \Rightarrow C = 2/9$.
Comparing coefficients of $(j\omega)^0$: $A(1)(4) + B(4) + C(1) = 2 \Rightarrow 4A + 8/3 + 2/9 = 2 \Rightarrow A = -2/9$.
Step 4: Inverse FT.
Using the known pairs $\frac{1}{a+j\omega} \leftrightarrow e^{-at}u(t)$ and $\frac{1}{(a+j\omega)^2} \leftrightarrow te^{-at}u(t)$:

$$\boxed{y(t) = \left(-\frac{2}{9}e^{-t} + \frac{2}{3}te^{-t} + \frac{2}{9}e^{-4t}\right)u(t)}$$

Checks:

- $y(0) = -2/9 + 0 + 2/9 = 0$ ✓ (starts at zero)
- $y(\infty) = 0$ ✓ (all exponentials decay)
- $y'(0) = 2/9 + 2/3 - 8/9 = 2/9 + 6/9 - 8/9 = 0$ ✓ (initial conditions are zero for an initially relaxed system)

15.7 Summary of the Four-Lecture Block

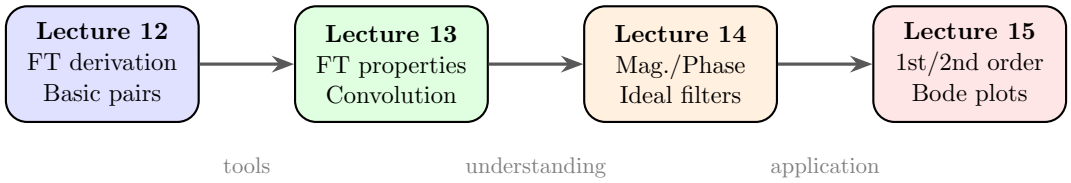

Figure 7: The progression across the four-lecture FT block: from derivation (12) through properties (13) to understanding (14) and finally application (15).

	First-Order	**Second-Order**
CT standard form	$H = \dfrac{1}{1 + j\omega/\omega_c}$	$H = \dfrac{1}{1 - (\omega/\omega_n)^2 + j2\zeta\omega/\omega_n}$
CT high-freq. rolloff	-20 dB/decade	-40 dB/decade
CT key parameter(s)	ω_c (cutoff)	ω_n (natural freq.), ζ (damping)
CT resonance?	No	Yes, when $\zeta < 1/\sqrt{2}$
DT standard form	$H = \dfrac{1-a}{1 - ae^{-j\omega}}$	Complex poles at $z = re^{\pm j\theta}$
DT stability	$\lvert a\rvert < 1$	$r < 1$
DT resonance freq.	N/A	$\omega \approx \theta$

15.8 Common Mistakes to Avoid

1. **Confusing ω_n with ω_c:** The natural frequency ω_n is **not** the -3 dB cutoff in a second-order system (unless $\zeta = 1/\sqrt{2}$). In general, the -3 dB bandwidth depends on both ω_n and ζ.

2. **Forgetting the dB/decade slope rules:** First-order $\rightarrow \pm 20$ dB/decade per pole or zero. Second-order $\rightarrow \pm 40$ dB/decade. Each additional pole adds -20 dB/decade; each additional zero adds $+20$ dB/decade.

3. **Plotting Bode plots on a linear frequency scale:** Bode plots **must** use a logarithmic frequency axis to be meaningful. On a linear scale, the asymptotic straight-line approximation does not work.

4. **Assuming $\zeta < 1$ always gives a resonance peak in $|H|$:** The magnitude peak exists only when $\zeta < 1/\sqrt{2} \approx 0.707$. For $1/\sqrt{2} < \zeta < 1$, the system is underdamped (oscillatory step response) but has **no** resonance peak in the magnitude response.

5. **Confusing DT pole radius with damping direction:** In DT, larger r means **less** damping (poles closer to the unit circle). In CT, poles farther from the imaginary axis (larger σ) mean **more** damping. The directions are "opposite" because stability boundaries differ (unit circle vs. imaginary axis).

6. **Not checking units in physical problems:** In the suspension example, $\omega_n = \sqrt{k/m}$ has units of $\sqrt{\mathrm{N/m/kg}} = \sqrt{\mathrm{s}^{-2}} = \mathrm{rad/s}$. Always verify dimensional consistency.

Key Insight

The central message of this lecture and the four-lecture block: The Fourier transform converts differential and difference equations into algebraic equations, convolution into multiplication, and system analysis into reading a Bode plot. First- and second-order systems are the building blocks: most practical systems can be decomposed into cascades and combinations of these standard forms. Master these, and you can analyze almost anything.

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